Belief State Geometry in Non-Ergodic Transformers: A Mess3 Mixture Experiment

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1 Motivation

1.1 Why Non-Ergodic Mess3 Training?

Real language corpora are *non-ergodic*: each document belongs to one domain, author, or style—and crucially, it never switches mid-document. A model trained on such data must simultaneously (a) identify *which* type of document it is reading (meta-belief π) and (b) track fine-grained structure within that document type (within-component belief η_k). These two tasks are epistemically distinct but must be solved together.

We formalise this with $K = 3$ **Mess3 HMMs**—3-state, 3-token edge-emitting Hidden Markov Models—where each training sequence is generated entirely by one component (non-ergodic corpus construction). This yields a clean, analytically tractable proxy for the more complex real-world situation.

Connection to in-context learning (ICL). Riechers et al. [2] prove that ICL emerges *necessarily* from non-ergodic pretraining: a model trained on a mixture of sources must learn to identify which source is generating the current sequence, and its in-context loss decay is governed by how quickly it can do so. They introduce the synchronisation horizon

$$N^* = \frac{1}{\min_{k \neq k'} D_{\mathrm{KL}}(P_k \| P_{k'})}$$

as the number of tokens required to commit to one component, and predict a sharp transition in in-context performance around position N^*. Their framework is information-theoretic and architecture-independent—it describes *what* must happen but not *where* in the model it happens. Our experiment provides the geometric complement: by tracking the meta-belief π directly in the residual stream, we can observe the synchronisation transition as a spatial phenomenon—a sharp rise in $R^2(\pi)$ near position N^*—and identify which subspace implements it. By choosing parameters with $N^* \approx 9.6 < 16$ (context window), this commitment happens within a single sequence.

Connection to factored representations. Shai et al. [3] show that transformers trained on parallel joint processes learn orthogonal subspaces in the residual stream—one per factor. Our experiment tests whether the same phenomenon occurs for *temporal* mixtures (one component per sequence, not simultaneous components).

The gap in the literature. Three closely related papers cover adjacent setups:

- Shai et al. [1]: single ergodic Mess3 $\rightarrow$ fractal geometry in residual stream.
- Riechers et al. [2]: non-ergodic mixture $\rightarrow$ ICL loss dynamics (but not geometry).
- Shai et al. [3]: parallel factors $\rightarrow$ factored geometry (but different setup).

This experiment fills the missing cell: *non-ergodic mixture $\rightarrow$ factored geometry?*

2 Setup

2.1 Data

Mess3 HMM. A Mess3 process is a 3-state, 3-token edge-emitting Hidden Markov Model parameterised by emission strength $\alpha \in (0.5, 1)$ and mixing rate $x \in (0, 1/3)$. The *token-labeled transition matrix* $T^{(a)} \in \mathbb{R}^{3\times 3}$ encodes the joint probability of emitting token a and transitioning to the next state:

$$T^{(a)}_{ij} = P(\text{token} = a \mid s_t = i) \cdot P(s_{t+1} = j \mid s_t = i),$$

where $P(\text{token} = a \mid s = i) = \alpha$ if $a = i$, else $(1-\alpha)/2$, and $P(s' = j \mid s = i) = 1 - 2x$ if $i = j$, else x. Sequences are generated by starting in a state drawn from the (uniform) stationary distribution, then at each step emitting a token according to the current state and transitioning to the next state jointly.

Components. We use three Mess3 HMMs with the following parameters:

Parameters were chosen so that: (a) components are distinguishable within 16 tokens ($N^* \approx 9.6 < 16$), and (b) each component produces a distinct fractal geometry in belief space. Figure 1 shows the

Component	α	x	Character
A	0.95	0.02	Sharp fractal, slow mixing
B	0.80	0.08	Medium fractal, moderate mixing
C	0.55	0.25	Sparse attractor, fastest mixing

Table 1: Mess3 component parameters. α is emission strength; x is the off-diagonal transition probability.

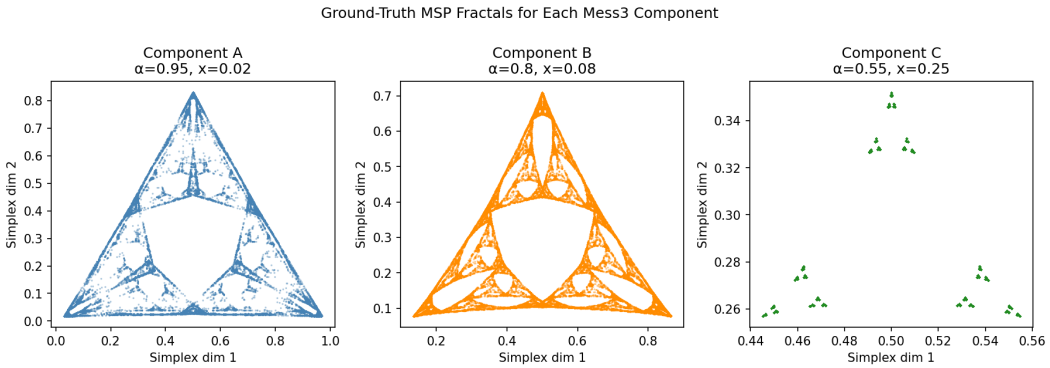

Figure 1: Ground-truth MSP fractals for each Mess3 component, simulated by iterating the IFS $f_a(\eta) = \eta T^{(a)}/(\eta T^{(a)}\mathbf{1})$ for 80,000 steps. Component A (blue) produces a rich self-similar fractal; component C (green) has only ∼9 reachable belief states due to weak emission ($\alpha = 0.55$).

Mixed State Presentation (MSP) attractor for each component—the fractal attractor of the belief state iterated function system (IFS).

Sequence generation. To form the training corpus, a component $k \sim \text{Uniform}(\{A, B, C\})$ is sampled for each sequence. A token sequence $x_{1:L}$ of length $L = 16$ (the context window) is then generated from component k alone, starting from the stationary distribution. The context window was set to $L = 16$ to satisfy $N^* < L$: with $N^* \approx 9.6$, the model accumulates enough context to synchronise to the correct component within a single sequence. The corpus is *non-ergodic* because sequences never switch components — each sequence is a pure draw from one Mess3 process. Figure 2 shows one example sequence per component; the visual contrast in token diversity across rows reflects the very different α and x values.

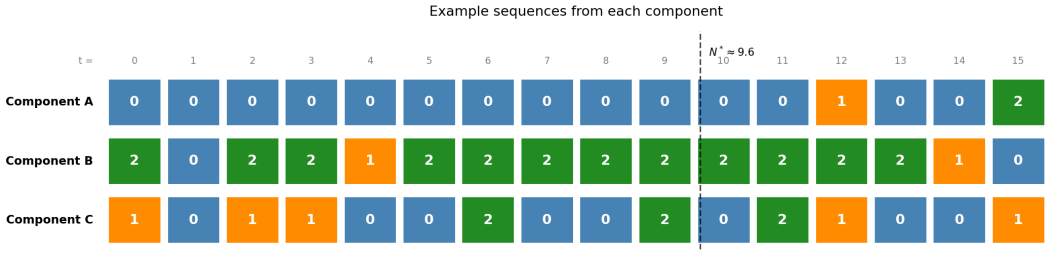

Figure 2: One example sequence (length 16) from each component. Cell color encodes token value (blue = 0, orange = 1, green = 2). Component A ($\alpha = 0.95$) is nearly always token 0; component C ($\alpha = 0.55$) mixes all three tokens freely. The dashed line marks the synchronisation horizon $N^* \approx 9.6$: before this point the identity of the component is ambiguous; after it a Bayesian observer has committed.

Ground-truth belief states. At each position t, the Bayesian-optimal agent maintains a joint belief $(\pi(t), \eta_A(t), \eta_B(t), \eta_C(t))$, where $\pi(t) \in \Delta^2$ is the meta-belief over components and $\eta_k(t) \in \Delta^2$ is the within-component belief for component k. Both are initialised to the uniform distribution and updated after each observed token using the two-step rule (Section 3.1). These analytically

computed belief states are stored alongside each sequence and used as regression targets in the geometry analysis. 50,000 training sequences and 5,000 validation sequences were generated.

Figure 3 illustrates the full joint 9-state belief geometry: the meta-belief π lives on the central simplex, while each vertex hosts the MSP fractal of the corresponding component. The three fractals are disconnected, reflecting the non-ergodic block structure.

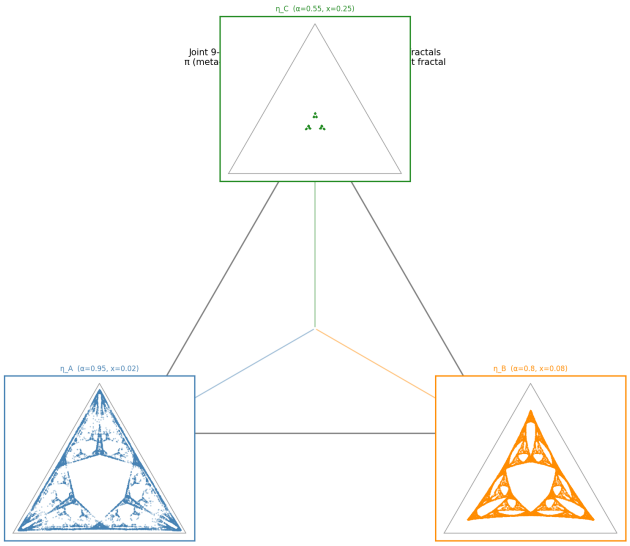

Figure 3: Joint 9-state belief geometry. The centre shows the meta-belief π simplex (which component?); each vertex hosts the MSP fractal attractor of that component (where within?). The three fractals are disconnected—no probability mass flows between components—reflecting the non-ergodic block structure of the corpus.

2.2 Architecture

Small GPT-2-style transformer via TransformerLens [4]: 2 layers, $d_{\text{model}} = 64$, 4 heads, $d_{\text{head}} = 16$, $d_{\text{mlp}} = 256$, context window 16, vocabulary $\{0, 1, 2\}$, $\approx$101k parameters.

2.3 Training

Cross-entropy next-token prediction, AdamW (lr $= 10^{-3}$, cosine decay to 10^{-5}, 1,000 warmup steps), 50,000 steps total. Sequences are generated on-the-fly each step to prevent overfitting. Checkpoints saved at steps 100, 500, 1,000, 2,000, then every 5,000 steps; at each checkpoint we run regression at the final layer and final position to track how R^2 evolves over training.

3 Hypotheses and Predictions

3.1 Mathematical Derivation

The joint belief state. The joint hidden state space is $\mathcal{S}_{\text{joint}} = \mathcal{S}_A \sqcup \mathcal{S}_B \sqcup \mathcal{S}_C$ (9 states). The joint token-labeled transition matrices are block-diagonal:

$$T^{(x)}_{\text{joint}} = \text{diag}\left(T^{(x)}_A,\, T^{(x)}_B,\, T^{(x)}_C\right).$$

The zeros off the block diagonal encode non-ergodicity exactly: no probability mass can flow between components. This block structure means the joint belief state $\eta_{\text{joint}} \in \Delta^8$ decomposes uniquely as

$$\eta_{\text{joint}}[k, s] = \pi_k \cdot \eta_k[s],$$

where $\pi \in \Delta^2$ is the *meta-belief* and $\eta_k \in \Delta^2$ is the *within-component belief*.

The two-step update. Upon observing token x:

$$L_k(x) = \eta_k \cdot T_k^{(x)} \cdot \mathbf{1} \quad \text{(per-component likelihood)}, \tag{1}$$

$$\pi'_k = \frac{\pi_k \cdot L_k(x)}{\sum_j \pi_j L_j(x)} \quad \text{(meta-belief Bayes update)}, \tag{2}$$

$$\eta'_k = \frac{\eta_k \cdot T_k^{(x)}}{L_k(x)} \quad \text{(within-component Bayes update)}. \tag{3}$$

Crucially, equation (3) applies the standard single-Mess3 update *independently* to each component. The components couple only through the normalisation constant

$$Z \;=\; \sum_j \pi_j \, L_j(x) \;=\; P(x_t = x),$$

which is the marginal probability of the observed token under the mixture, in equation (2).

Conditional independence. Conditioned on component identity k, the within-component beliefs η_j for $j \neq k$ evolve independently of η_k. This is the condition under which Shai et al. [3] (Theorem 1) proves that factored representations are lossless: no information is lost by encoding π and each η_k in separate orthogonal subspaces.

Dimensionality prediction. The full belief space is 8-dimensional ($\dim \Delta^8 = 8$). Under H2 (factored), PCA should reveal 8 principal components with non-negligible variance, structured as 4 pairs: a 2D π subspace (simplex, not fractal) and three 2D η_k subspaces (each a Mess3 fractal).

Synchronisation horizon. At the start of a sequence the reader has no information about which component is generating tokens: $\pi(0) = (1/3, 1/3, 1/3)$. Each sequence is generated by a single component $k \sim \mathrm{Uniform}(\{A, B, C\})$, but the component identity is hidden—only the tokens are observed. Each observed token provides a small amount of evidence—component A mostly emits token 0, while component C mixes all three freely—so $\pi(t)$ gradually sharpens toward one vertex of the simplex. More precisely, Bayes' rule updates $\pi(t)$ by multiplying each component's weight by its likelihood $L_k(x)$, so components whose token statistics better match the observed tokens gain weight while the others decay. The *synchronisation horizon* N^* is the characteristic number of tokens required for $\pi(t)$ to collapse near the vertex of the true component—i.e. for the reader to have effectively identified which component generated this sequence.

Crucially, collapse only happens because the components have *different* token statistics. If all components were identical, every token would carry equal likelihood under every component ($L_k(x) = L_{k'}(x)$ for all k, k'), the ratio $\pi_k(t)/\pi_{k'}(t)$ would never change, and π would remain uniform forever ($N^* \to \infty$). N^* is therefore a measure of how distinguishable the components are: the more similar they are per token (small D_{KL}), the longer identification takes. The hardest-to-distinguish pair is the bottleneck, giving

$$N^* = \frac{1}{\min_{k \neq k'} D_{\mathrm{KL}}(P_k \| P_{k'})},$$

where D_{KL} is estimated empirically via per-token log-likelihood ratios over long sequences. With the chosen parameters, $N^* \approx 9.6 < 16$, so synchronisation completes within every training sequence.

3.2 Competing Hypotheses

H1 (Joint): The transformer represents η_{joint} as one undifferentiated 8D blob. High R^2 for all targets, but projecting out the π subspace substantially degrades η_k regression.

H2 (Factored): Four orthogonal subspaces: one 2D π subspace and three 2D η_k subspaces. Projecting out π leaves η_k $R^2 \approx$ unchanged.

H3 (Superposition): π and η_k recoverable approximately but packed into fewer than 8 effective dimensions via near-orthogonal superposition.

3.3 Intuitive Prediction

My geometric intuition is that the residual stream should mirror the ground-truth joint belief geometry: the *nested fractal* shown in Figure 3, with π occupying a central simplex and each vertex of that simplex hosting the MSP fractal attractor of the corresponding component.

I favoured H2 (Factored) by analogy to Shai et al. [3], where the setting is a *parallel* joint process—at each timestep independent subprocesses each emit a token and the observation is their concatenation. Our setup is *sequential*—each sequence comes from one subprocess—but the underlying math is the same: block-diagonal transition matrices, conditionally independent factors, and a belief space that decomposes into orthogonal subspaces. If transformers learn factored representations in the parallel case, I expected them to do so here too.

My one pre-registered quantitative prediction was that **CEV dimensionality** ≈ 8 in the final layer, corresponding to four 2D subspaces under H2.

4 Methods

PCA and CEV dimensionality. To measure how many dimensions the transformer uses to represent the belief state, we run PCA on the matrix of all residual stream vectors per layer, flattening sequences and positions into rows to obtain an $(N \cdot L) \times d_{\text{model}}$ matrix. The principal components are ordered by variance explained. The **CEV (cumulative explained variance) dimensionality** at threshold p is the smallest d such that the top-d components together explain $\geq p\%$ of total variance—the minimum number of directions needed to capture $p\%$ of the structure in the activations.

Regression-based subspace identification [1]. We fit a linear map $W \in \mathbb{R}^{3 \times d_{\text{model}}}$ from residual stream activations $\mathbf{a} \in \mathbb{R}^{d_{\text{model}}}$ (normalised to zero mean and unit variance per dimension) to ground-truth belief targets $\mathbf{b} \in \mathbb{R}^3$ (either π or η_k) via Ridge regression ($\alpha = 10^{-4}$). Linear decodability is measured by the coefficient of determination pooled across all three simplex coordinates,

$$R^2 = 1 - \frac{\sum_{i,j}(b_{ij} - \hat{b}_{ij})^2}{\sum_{i,j}(b_{ij} - \bar{b}_j)^2},$$

reported on a held-out 20% test split (fit on 80%). To identify the subspace encoding the belief, we compute the SVD $W = U\Sigma V^\top$: the top-2 right singular vectors (rows of $V^\top$) are the 2 directions in activation space along which the predictor varies most, spanning the dominant 2D belief-encoding subspace. This basis is used in all subsequent overlap tests.

Subspace overlap metric [3]. Given two subspaces with orthonormal bases $Q_A \in \mathbb{R}^{k_A \times d}$ and $Q_B \in \mathbb{R}^{k_B \times d}$ (orthonormal rows), the overlap is

$$\text{overlap}(Q_A, Q_B) = \frac{\|Q_A Q_B^\top\|_F}{\sqrt{k_A \cdot k_B}}.$$

The (i, j) entry of $Q_A Q_B^\top$ is the cosine $\cos\theta_{ij}$ between the i-th basis vector of Q_A and the j-th basis vector of Q_B; the Frobenius norm sums their squares, and dividing by $\sqrt{k_A k_B}$ normalises to $[0, 1]$ (0 = orthogonal subspaces, 1 = identical subspaces).

Fractal visualisation. We visualise whether the residual stream encodes the full fractal geometry of each component's belief attractor, not merely a summary statistic. For each component k, we take residual stream activations from component-k sequences at positions $t > N^*$ (after the meta-belief π has collapsed, so activations reflect pure component-k belief states), apply the η_k regression map to obtain predicted $\hat{\eta}_k \in \mathbb{R}^3$, and plot the resulting point cloud in barycentric simplex coordinates. We compare this cloud against the ground-truth MSP attractor, generated by iterating the IFS $f_a(\eta) = \eta T^{(a)} / (\eta T^{(a)} \mathbf{1})$ for 80,000 steps.

5 Results

5.1 PCA Dimensionality

The table reveals a striking compression: the transformer uses fewer dimensions than the theoretical belief space requires. Theory predicts 8 dimensions (four 2D subspaces), yet 98% of variance is captured in just 7 dimensions at the final layer—and 95% in only 5.

This suggests the transformer is *more efficient* than the theoretical minimum representation. We offer three possible explanations, though we cannot confirm which dominates:

Layer	CEV (95%)	CEV (98%)	CEV (100%)	Predicted
0	6	9	64	8
1	5	7	64	8

Table 2: PCA dimensionality per layer. CEV($p\%$) = fewest components explaining $\geq p\%$ of variance. The first 5–6 dims capture 95%, 7–9 capture 98%, and all 64 carry non-zero variance, revealing a low-dimensional dominant subspace atop a long noise tail.

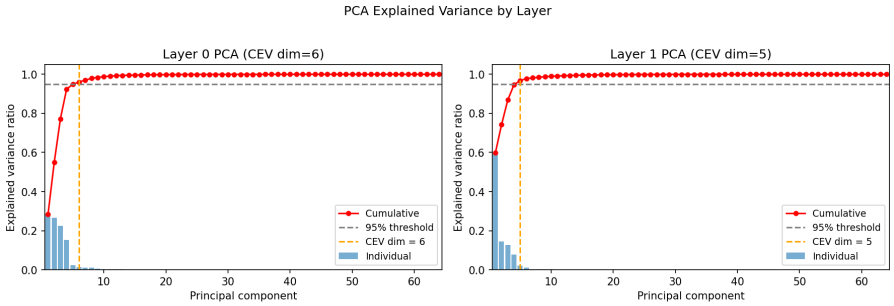

Figure 4: Cumulative explained variance by layer. Layer 1 (final) uses 5 components to explain 95% of variance.

1. **Component C's near-degenerate attractor.** With $\alpha = 0.55$, component C has only $\sim$9 reachable belief states, collapsing η_C from a 2D subspace toward a near-discrete structure that requires far less than 2 dimensions.
2. **Fractal attractors don't fill their 2D subspaces uniformly.** Each MSP attractor is a fractal with dimension strictly less than 2. Belief states cluster along lower-dimensional structures within each 2D subspace, so PCA sees less than 2 effective dimensions per factor.
3. **Superposition enabled by attractor sparsity.** Because the attractors are sparse, the transformer can pack multiple belief directions into shared dimensions without interference—achieving near-perfect $R^2 > 0.95$ with fewer dimensions than a uniform distribution would require.

At 100% CEV both layers require all 64 dimensions, confirming a sharp transition between the dominant low-dimensional belief subspace and a long tail of noise dimensions.

5.2 Linear Decodability

Quantity	R^2 (in-sample)	R^2 (out-of-sample)
π (meta-belief)	0.956	0.957
η_A (component A)	0.950	0.950
η_B (component B)	0.964	0.963
η_C (component C)	0.985	0.984

Table 3: Ridge regression R^2 at the final layer, final context position (position 15). In-sample and out-of-sample (80/20 split) R^2 are nearly identical, confirming no overfitting.

All belief targets are linearly decodable with $R^2 > 0.95$, confirming that the residual stream encodes the full joint belief state. In-sample and out-of-sample R^2 are nearly identical, ruling out overfitting.

Figure 5 shows R^2 as a function of context position. $R^2(\eta_k)$ is high from the very first position in both layers—the within-component belief is encoded reliably regardless of how much context has accumulated. $R^2(\pi)$ tells a different story across layers: in Layer 0 it peaks early then decays sharply, while in Layer 1 it remains high throughout. This suggests that Layer 0 encodes the meta-belief primarily from the most recent token (useful but not persistent), while Layer 1 maintains a stable π representation that accumulates over the full context. The aggregate $R^2(\pi) = 0.956$ in Table 3 reflects this Layer 1 stability averaged over all positions.

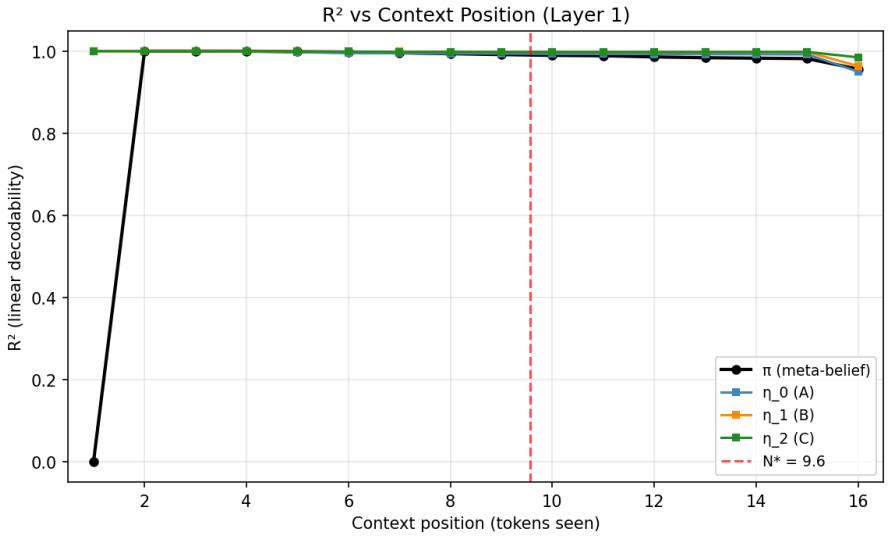

Figure 5: R^2 vs. context position for both layers. $R^2(\eta_k)$ (coloured) is high from position 1 in both layers. $R^2(\pi)$ (black) decays in Layer 0 but remains stable in Layer 1, indicating that meta-belief is persistently encoded only in the final layer. The red dashed line marks $N^* \approx 9.6$.

5.3 Subspace Orthogonality

	π	η_A	η_B	η_C
π	1.000	0.193	0.171	0.176
η_A	0.193	1.000	0.691	0.534
η_B	0.171	0.691	1.000	0.590
η_C	0.176	0.534	0.590	1.000

Table 4: Subspace overlap matrix (0 = orthogonal, 1 = identical). π is nearly orthogonal to all η_k subspaces (0.17–0.19). The η_k subspaces overlap each other moderately, reflecting the shared token alphabet.

π is nearly orthogonal to all η_k subspaces (overlap 0.17–0.19). The moderate η_k–$\eta_{k'}$ overlap (0.53–0.69) reflects the shared 3-token alphabet: all components respond to the same tokens, inducing partially shared within-component directions.

As a direct functional test, removing the π subspace from activations and re-running η_k regression yields R^2 drops of less than 0.001 in all cases, and removing any η_k subspace leaves π regression unchanged—confirming that the subspaces are functionally independent, not merely geometrically near-orthogonal.

5.4 Fractal Geometry

Figure 6 shows, for each component, the ground-truth belief state trajectories (left) and the predicted η_k obtained by applying the regression weights to residual stream activations (right), both plotted in barycentric simplex coordinates.

The regression output faithfully reproduces the fractal structure of each component's MSP attractor, confirming that the model encodes the full geometric structure of the belief distribution, not merely its mean. Component A exhibits a rich self-similar fractal; component B a moderately nested fractal; component C a sparse set of $\sim$9 discrete clusters. This confirms that the residual stream encodes not merely the mean of the belief distribution but the full geometric structure of the attractor.

5.5 Training Dynamics

Figure 7 and Table 5 show that η_k subspaces emerge *before* the π subspace.

Belief	Phase transition (step at $R^2 > 0.9$)
η_A	100
η_B	100
η_C	100
π (meta-belief)	2,000

Table 5: Training dynamics phase transitions.

The ordering—within-component beliefs first, meta-belief later—suggests that the transformer first learns to track the local token statistics of each component individually (which requires only learning the Mess3 update within each component's context), and only subsequently learns to *distinguish* which component is active (which requires comparing cross-component likelihoods). This ordering makes sense: per-component prediction reduces loss immediately, while improving π only helps once the model has already learned the within-component structure to compare.

6 Additional Analysis: Vary-One

Motivation. The vary-one analysis tests whether the learned η_k subspaces are *component-specific*. Under H2, each η_k should live in its own 2D subspace. Fitting the η_k regression on only component-k sequences (fixing component identity, varying within-component positions) reveals whether the model has learned three distinct fractal geometries or one shared one. If the cross-component subspace overlap is low, each component occupies a distinct region of residual stream space.

Results.

Component	Within-component η_k R^2
A	0.990
B	0.962
C	0.980

Table 6: Vary-one R^2: regression fit using only sequences from component k.

Within-component R^2 is high for all components (≥ 0.96, Table 6), confirming that sequences from the same component share a coherent subspace structure. For component A the single-component fit is notably better (0.990 vs. 0.950), while for B and C the difference is small, likely because their more uniform attractors are already well-characterised by the global fit.

	A	B	C
A	1.000	0.275	0.208
B	0.275	1.000	0.510
C	0.208	0.510	1.000

Table 7: Cross-component subspace overlap from vary-one analysis. A and C are nearly orthogonal (0.208); A–B and B–C show moderate overlap, likely due to similar emission statistics.

The cross-component overlaps (Table 7) are substantially lower than the η_k–$\eta_{k'}$ overlaps in the global analysis (Table 4), especially for A–C (0.208). This confirms that the transformer has learned *component-specific* fractal geometries. Component B's moderate overlap with both A and C likely reflects its intermediate emission strength ($\alpha = 0.80$), making its belief geometry somewhat "between" those of A and C.

7 Discussion

7.1 Which Hypothesis Best Fits?

The picture is mixed: H2 holds cleanly for the π–η split but not for the internal structure of the η_k subspaces.

H2 confirmed for π vs. η_k. The projection tests are decisive: removing the π subspace from activations causes R^2 drops of less than 0.001 for all η_k targets, and removing any η_k subspace leaves π regression unchanged. The π–η_k subspace overlaps (0.17–0.19) confirm near-orthogonality. The transformer has cleanly separated meta-belief from within-component belief into distinct residual stream directions.

H2 not confirmed for η_k vs. $\eta_{k'}$. H2 predicts four mutually orthogonal subspaces, but the three η_k subspaces overlap substantially with each other (0.53–0.69). This is more consistent with H3 (Superposition): the model does not dedicate a separate 2D subspace to each component's fractal, but instead packs the three within-component beliefs into a shared, partially overlapping region of residual stream space. The vary-one analysis shows that component-specific fits are more distinct (A–C overlap 0.21) than the global fits suggest, so the components are not fully merged—but they are far from orthogonal.

The pre-registered prediction (CEV dimensionality ≈ 8) was not confirmed: both layers use only 5–6 dimensions to explain 95% of variance, consistent with the compressed η_k representation described above.

A notable surprise in the training dynamics: within-component beliefs (η_k) emerge early (step 100, $R^2 > 0.9$), while the meta-belief (π) lags behind (step 2,000). This suggests the model first learns to track per-component token statistics (which reduces loss immediately) and only subsequently learns to distinguish *which* component is active (which requires cross-component comparison).

7.2 Implications

Non-ergodic pretraining induces partial factoring. The transformer cleanly separates "what component am I in?" (π) from "where within this component am I?" (η) without explicit supervision. However, the three within-component beliefs are not further separated from each other—they share residual stream directions, suggesting that factoring happens at the coarsest level of the hierarchy (meta vs. within) but not below it.

Two-level belief structure maps onto two-phase training dynamics. Within-component geometry is learned first and meta-belief routing emerges second—the ordering reflects loss reduction efficiency, not conceptual priority.

In-context learning has geometric structure. ICL in real language models may involve a π-like routing circuit that activates domain-specific η-like representations. The factored geometry found here suggests these circuits can be linearly separated in residual stream space.

7.3 Limitations

Small model ($d = 64$) and short context ($L = 16$) may limit generality. The symmetric 3-state/3-token structure makes components easier to distinguish than typical natural language domains. Ground-truth belief computation uses exact analytical updates—real corpora do not provide such ground truth.

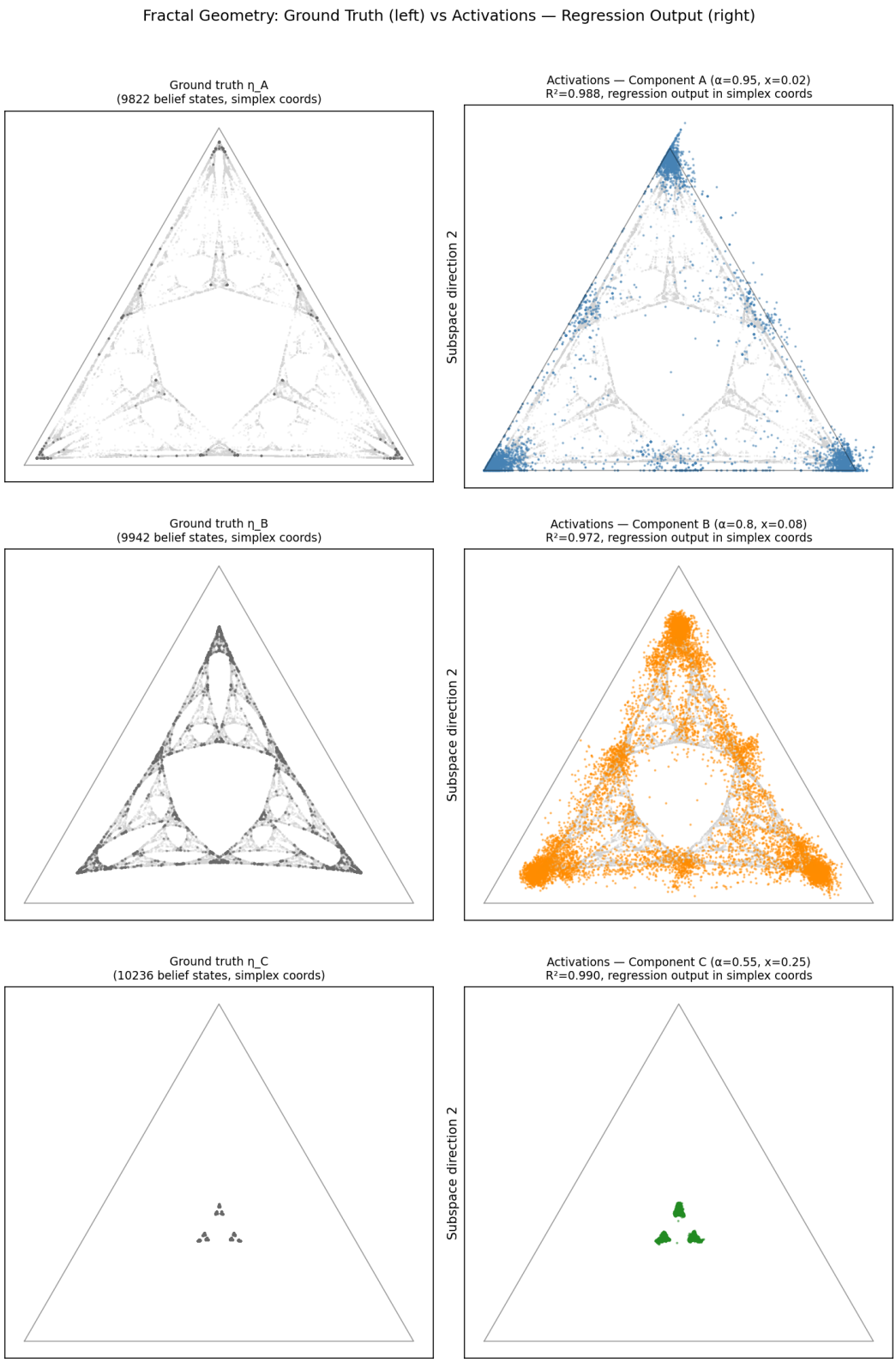

Figure 6: Fractal comparison for each component: ground-truth η_k trajectories (left) vs. regression-predicted η_k from residual stream activations (right), plotted in barycentric simplex coordinates. The regression output reproduces the MSP fractal geometry.

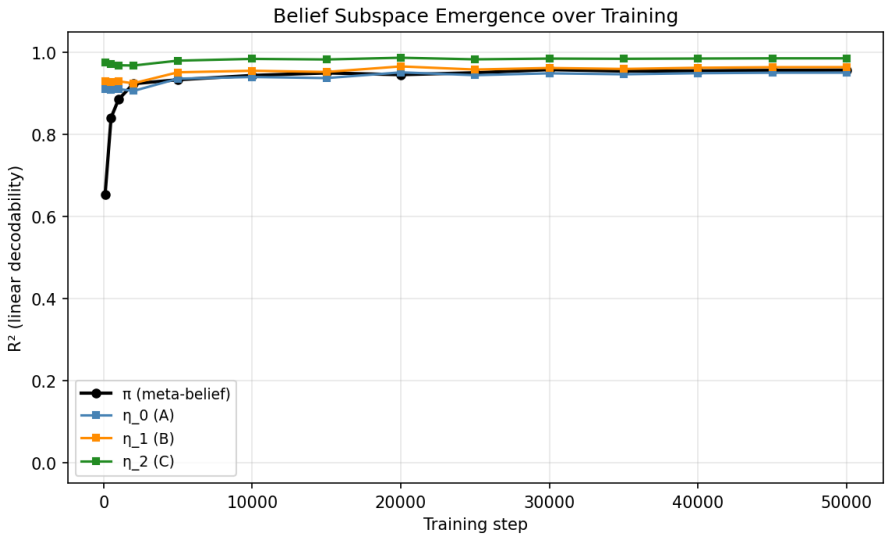

Figure 7: R^2 of π and η_k regressions over training steps (final layer, final position). η_k representations emerge early (step 100, $R^2 > 0.9$); π representation emerges later (step 2,000).

References

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